2.60 Graph Sketching: Advanced Rational Functions

Worked Solutions

1a)

Polynomial long division:

$$x^2 - x + 2 - (x)(x+1) = 2 - 2x,$$
$$2 - 2x - (-2)(x+1) = 4.$$

Polynomial division gives

$$\frac{x^2 - x + 2}{x+1} = x - 2 + \frac{4}{x+1}.$$

The asymptotes are $x = -1$ and $y = x - 2$.

1b)

Polynomial long division:

$$x^2 + 2x - 6 - (x)(x-4) = 6x - 6,$$
$$6x - 6 - (6)(x-4) = 18.$$

$$\frac{x^2 + 2x - 6}{x-4} = x + 6 + \frac{18}{x-4}.$$

The asymptotes are $x = 4$ and $y = x + 6$.

1c)

Polynomial long division:

$$3x^2 + x - 7 - (3x)(x+2) = -5x - 7,$$
$$-5x - 7 - (-5)(x+2) = 3.$$

$$\frac{3x^2 + x - 7}{x+2} = 3x - 5 + \frac{3}{x+2}.$$

The asymptotes are $x = -2$ and $y = 3x - 5$.

1d)

Polynomial long division:

$$11 - x^2 - (-x)(x+3) = 3x + 11,$$
$$3x + 11 - (3)(x+3) = 2.$$

$$\frac{11 - x^2}{x+3} = -x + 3 + \frac{2}{x+3}.$$

The asymptotes are $x = -3$ and $y = -x + 3$.

2a)

Polynomial long division:

$$x^2 - 2x + 1 - (x)(x+1) = 1 - 3x,$$
$$1 - 3x - (-3)(x+1) = 4.$$

From $y = x - 3 + \dfrac{4}{x+1}$, the asymptotes are $x = -1$ and $y = x - 3$. The intercepts are $(0, -1)$ and $(1 - \sqrt{2}, 0), (1 + \sqrt{2}, 0)$.

2b)

Polynomial long division:

$$2x^2 + 7x + 1 - (2x)(x+3) = x + 1,$$
$$x + 1 - (1)(x+3) = -2.$$

From $y = 2x + 1 - \dfrac{2}{x+3}$, the asymptotes are $x = -3$ and $y = 2x + 1$. The intercepts are $(0, \frac{1}{3})$ and $(\frac{-7-\sqrt{41}}{4}, 0), (\frac{-7+\sqrt{41}}{4}, 0)$.

2c)

Polynomial long division:

$$6x^2 + x + 6 - (-2x)(1-3x) = 3x + 6,$$
$$3x + 6 - (-1)(1-3x) = 7.$$

From $y = -2x - 1 + \dfrac{7}{1-3x}$, the asymptotes are $x = \frac{1}{3}$ and $y = -2x - 1$. The y-intercept is $(0, 6)$. The numerator $6x^2 + x + 6 = 0$ has discriminant $1 - 144 < 0$, so there are no x-intercepts.

2d)

Polynomial long division:

$$-x^2 + 5x + 51 - (-x)(x+5) = 10x + 51,$$
$$10x + 51 - (10)(x+5) = 1.$$

From $y = 10 - x + \dfrac{1}{x+5}$, the asymptotes are $x = -5$ and $y = 10 - x$. The y-intercept is $(0, \frac{51}{5})$. The x-intercepts satisfy $51 + 5x - x^2 = 0$, so $x = \dfrac{5 \pm \sqrt{229}}{2}$.

3a) The curve has vertical asymptote $x = 2$, oblique asymptote $y = 5 - 2x$ and passes through $(0, 0)$. Write $y = 5 - 2x + \dfrac{k}{x-2}$. Substituting $(0, 0)$ gives $k = 10$. Thus

$$y = \frac{5(x-2) - 2x(x-2) + 10}{x-2} = \frac{-2x^2 + 9x}{x-2},$$

so $(a, b, c, d) = (-2, 9, 0, -2)$.

3b) The asymptotes are $x = -2$ and $y = x - 1$, and the curve passes through $(0, \frac{1}{2})$. Writing $y = x - 1 + \dfrac{k}{x+2}$ gives $k = 3$, and

$$y = \frac{x(x+2) - (x+2) + 3}{x+2} = \frac{x^2 + x + 1}{x+2}.$$

Hence $(a, b, c, d) = (1, 1, 1, 2)$.

3c) The asymptotes are $x = -1$ and $y = 2 - x$, and the curve passes through $(0, 4)$. Thus $y = 2 - x + \dfrac{2}{x+1}$ and

$$y = \frac{(2-x)(x+1) + 2}{x+1} = \frac{-x^2 + x + 4}{x+1},$$

so $(a, b, c, d) = (-1, 1, 4, 1)$.

3d) The asymptotes are $x = 1$ and $y = 2x + 1$, and the curve passes through $(0, -1)$. Thus $y = 2x + 1 + \dfrac{2}{x-1}$ and

$$y = \frac{(2x+1)(x-1)+2}{x-1} = \frac{2x^2 - x + 1}{x-1},$$

so $(a, b, c, d) = (2, -1, 1, -1)$.

4a)

Polynomial long division:

$$ax^2 - (ax)(x+1) = -ax,$$
$$-ax - (-a)(x+1) = a.$$

$$\frac{ax^2}{x+1} = ax - a + \frac{a}{x+1}.$$

The asymptotes are $x = -1$ and $y = ax - a$.

4b)

Polynomial long division:

$$ax^2 - (ax)(x-2) = 2ax,$$
$$2ax - (2a)(x-2) = 4a.$$

$$\frac{ax^2}{x-2} = ax + 2a + \frac{4a}{x-2}.$$

The asymptotes are $x = 2$ and $y = ax + 2a$.

4c)

Polynomial long division:

$$ax^2 + bx - (ax)(x-1) = ax + bx,$$
$$ax + bx - (a+b)(x-1) = a + b.$$

$$\frac{ax^2 + bx}{x-1} = ax + a + b + \frac{a+b}{x-1}.$$

The asymptotes are $x = 1$ and $y = ax + a + b$.

4d)

Polynomial long division:

$$ax^2 + b - (ax)(a+x) = -a^2x + b,$$
$$-a^2x + b - \left(-a^2\right)(a+x) = a^3 + b.$$

$$\frac{ax^2 + b}{x+a} = ax - a^2 + \frac{a^3 + b}{x+a}.$$

The asymptotes are $x = -a$ and $y = ax - a^2$.

4e)

Polynomial long division:

$$ax^2 - \left(\frac{ax}{b}\right)(bx - c) = \frac{acx}{b},$$
$$\frac{acx}{b} - \left(\frac{ac}{b^2}\right)(bx - c) = \frac{ac^2}{b^2}.$$

$$\frac{ax^2}{bx - c} = \frac{a}{b}x + \frac{ac}{b^2} + \frac{ac^2}{b^2(bx - c)}.$$

The asymptotes are $x = \frac{c}{b}$ and $y = \frac{a}{b}x + \frac{ac}{b^2}$.

5a) The other asymptote is the vertical asymptote $x = 1$.

5b)

Polynomial long division:

$$kx + x^2 + 4 - (x)(x - 1) = kx + x + 4,$$
$$kx + x + 4 - (k + 1)(x - 1) = k + 5.$$

Division gives $y = x + (k + 1) + \dfrac{k + 1}{x - 1}$. Since the oblique asymptote is $y = x + 3$, $k + 1 = 3$, so $k = 2$.

6a) The other asymptote is the vertical asymptote $x = 2$.

6b)

Polynomial long division:

$$kx + 2x^2 - 3 - (2x)(x - 2) = kx + 4x - 3,$$
$$kx + 4x - 3 - (k + 4)(x - 2) = 2k + 5.$$

Division gives $y = 2x + (k + 4) + \dfrac{2k + 5}{x - 2}$. Thus $k + 4 = 7$ and $k = 3$.

7a)

Polynomial long division:

$$ax^2 + bx + 4 - (ax)(x - 1) = ax + bx + 4,$$
$$ax + bx + 4 - (a + b)(x - 1) = a + b + 4.$$

Division gives $y = ax + (a + b) + \dfrac{a + b + 4}{x - 1}$. Hence $a = 3$ and $a + b = 2$, so $b = -1$.

7b) The other asymptote is $x = 1$.

8a) Put $f(x) = \dfrac{x^2 + 4x + 7}{x + 1}$. Setting $f(x) = k$ gives

$$x^2 + (4 - k)x + (7 - k) = 0.$$

For real intersections its discriminant is non-negative:

$$(4 - k)^2 - 4(7 - k) = (k + 2)(k - 6) \geq 0.$$

8b) At a stationary point the line $y = k$ is tangent, so the discriminant is zero: $k = -2$ or $k = 6$. Substitution gives stationary points $(-3, -1)$ and $(1, 7)$.

8c) Division gives $f(x) = x + 3 + \dfrac{4}{x+1}$. The asymptotes are $x = -1$ and $y = x + 3$.

8d) The curve has a local maximum at $(-3, -1)$ and a local minimum at $(1, 7)$; it approaches the asymptotes $x = -1$ and $y = x + 3$.

9a) Setting $f(x) = k$ gives

$$-x^2 + (6 - k)x + (12 - 2k) = 0.$$

The discriminant condition simplifies to $(k - 6)(k - 14) \geq 0$.

9b) Equality occurs at $k = 6, 14$. Substitution gives stationary points $(0, 6)$ and $(-4, 14)$.

9c) Division gives $f(x) = -x + 8 - \dfrac{4}{x+2}$. The asymptotes are $x = -2$ and $y = -x + 8$.

9d) The curve has a local minimum at $(0, 6)$ and a local maximum at $(-4, 14)$.

10a) For $t \geq 0$, the numerator $(2t - 3)(t + 1)$ is negative for $0 \leq t < \frac{3}{2}$ and positive for $t > \frac{3}{2}$. The rate curve crosses the t-axis at $t = \frac{3}{2}$ and has vertical asymptote $t = -2$ outside the domain.

10b) The population decreases until $t = \frac{3}{2}$ years, then increases.

10c) Division and integration give

$$\frac{dP}{dt} = 2t - 5 + \frac{7}{t+2}, \qquad P(t) = t^2 - 5t + 7\ln(t + 2) + C.$$

10d) From $P(0) = 4$, $C = 4 - 7\ln 2$. Therefore $P(t) = t^2 - 5t + 7\ln(t + 2) + 4 - 7\ln 2$.

10e) $P(4) = 7\ln 3 \approx 7.69$ thousand residents.

11a)

Polynomial long division:

$$ax^2 + bx + 7 - (ax)(x + 3) = -3ax + bx + 7,$$
$$-3ax + bx + 7 - (-3a + b)(x + 3) = 9a - 3b + 7.$$

Division gives $y = ax + (b - 3a) + \dfrac{7 - 3b + 9a}{x+3}$. Comparing with $y = x + 1$ gives $a = 1$ and $b = 4$.

11b) The other asymptote is $x = -3$.

11c) The line $y = k$ intersects when

$$x^2 + (4 - k)x + (7 - 3k) = 0.$$

At a stationary point the discriminant is zero:

$$(4 - k)^2 - 4(7 - 3k) = (k + 6)(k - 2) = 0.$$

For $k = -6$, $x = -5$; for $k = 2$, $x = -1$. The stationary points are $(-5, -6)$ and $(-1, 2)$.

11d) The curve has a local maximum at $(-5,-6)$ and a local minimum at $(-1,2)$, with asymptotes $x=-3$ and $y=x+1$.

12a)

Polynomial long division:

$$x^2-3-(x)\,(x+2)=-2x-3,$$
$$-2x-3-(-2)\,(x+2)=1.$$

Division gives $y=x-2+\dfrac{4}{x+2}$, so the asymptotes are $x=-2$ and $y=x-2$.

12b) Setting $y=k$ gives $(x+2)^2-(k+4)(x+2)+4=0$. The discriminant is zero when $k=0$ or $k=-8$. Substitution gives stationary points $(0,0)$ and $(-4,-8)$.

12c) The curve has a local maximum at $(-4,-8)$ and a local minimum at $(0,0)$.

12d) Using $y=x-2+\dfrac{4}{x+2}$,

$$V=\pi\int_0^2\frac{x^4}{(x+2)^2}\,dx=\pi\int_2^4\left(u^2-8u+24-\frac{32}{u}+\frac{16}{u^2}\right)du=\pi\left(\frac{68}{3}-32\ln 2\right),\quad u=x+2.$$

13a)

Polynomial long division:

$$3x^2+4x+2-(3x)\,(x+1)=x+2,$$
$$x+2-(1)\,(x+1)=1.$$

Division gives $C=3x+1+\dfrac{1}{x+1}$. The asymptotes are $x=-1$ and $C=3x+1$.

13b) Setting $C=k$ gives $3x^2+(4-k)x+(2-k)=0$. A repeated root requires $(4-k)^2-12(2-k)=0$, so $k=-2\pm 2\sqrt{3}$. The corresponding x-coordinates are $-1\pm\frac{1}{\sqrt{3}}$. To three decimal places the stationary points are $(-1.577,-5.464)$ and $(-0.423,1.464)$.

13c) The curve has a local maximum at $(-1.577,-5.464)$ and a local minimum at $(-0.423,1.464)$, with asymptotes $x=-1$ and $C=3x+1$.

13d) The mean value is

$$\frac{1}{4-1}\int_1^4\left(3x+1+\frac{1}{x+1}\right)dx=\frac{17}{2}+\frac{1}{3}\ln\frac{5}{2}\approx 8.805.$$

13e) The average production cost over the interval from 1 to 4 batches per day is approximately 8.805 hundreds of pounds per batch.

14a)

Polynomial long division:

$$x^3 - 4 - \left(x^2\right)(x+2) = -2x^2 - 4,$$
$$-2x^2 - 4 - (-2x)(x+2) = 4x - 4,$$
$$4x - 4 - (4)(x+2) = -12.$$

$\dfrac{x^3-4}{x+2} = x^2 - 2x + 4 - \dfrac{12}{x+2}$; the asymptotes are $x = -2$ and $y = x^2 - 2x + 4$.

14b)

Polynomial long division:

$$x^3 + 3x^2 + 1 - \left(x^2\right)(x-2) = 5x^2 + 1,$$
$$5x^2 + 1 - (5x)(x-2) = 10x + 1,$$
$$10x + 1 - (10)(x-2) = 21.$$

$\dfrac{x^3+3x^2+1}{x-2} = x^2 + 5x + 10 + \dfrac{21}{x-2}$; the asymptotes are $x = 2$ and $y = x^2 + 5x + 10$.

14c)

Polynomial long division:

$$1 - x^3 - \left(-x^2\right)(x+1) = x^2 + 1,$$
$$x^2 + 1 - (x)(x+1) = 1 - x,$$
$$1 - x - (-1)(x+1) = 2.$$

$\dfrac{1-x^3}{x+1} = -x^2 + x - 1 + \dfrac{2}{x+1}$; the asymptotes are $x = -1$ and $y = -x^2 + x - 1$.

15a) The printed solution works with numerator x^3+x^2+4: $y = x^2 + \dfrac{4}{x+1}$. It gives asymptotes $x = -1$ and $y = x^2$, y-intercept $(0, 4)$, and no real x-intercept. The question prints x^3+x^2-4; that wording is retained in the question document.

15b) $y = -x^2 + 1 + \dfrac{15}{x+3}$. The asymptotes are $x = -3$ and $y = -x^2 + 1$; the y-intercept is $(0, 6)$, and the only real x-intercept is $x = 2$.

15c) $y = x^2 - 4x + \dfrac{12}{x+1}$. The asymptotes are $x = -1$ and $y = x^2 - 4x$; the y-intercept is $(0, 12)$ and the x-intercepts are $-2, 2, 3$.

15d) The printed solution works with numerator $3x^3+4x^2+4x+1$: $y = 3x^2 + 10x + 24 + \dfrac{49}{x-2}$. The asymptotes are $x = 2$ and $y = 3x^2 + 10x + 24$; the y-intercept is $(0, -\frac{1}{2})$ and the only real x-intercept is $-\frac{1}{3}$. The question prints $3x^3 + 4x^2 + x + 1$; that wording is retained in the question document.

16a) The graph has asymptotes $x = 3$ and $y = x^2 + 3x + 11$, and y-intercept $\frac{7}{3}$. Write $y = x^2 + 3x + 11 + \dfrac{k}{x-3}$. Substitution of $x = 0$ gives $k = 26$. Therefore $y = \dfrac{x^3 + 2x - 7}{x-3}$, so $a = 1$, $b = 0$, $c = 2$, $d = -7$, and $p = -3$.

16b) The graph has asymptotes $x = -1$ and $y = -x^2 - x + 1$, and y-intercept 3. Thus $y = -x^2 - x + 1 + \dfrac{2}{x+1}$, so $y = \dfrac{-x^3 - 2x^2 + 3x + 1}{x+1}$. Hence $a = -1$, $b = -2$, $c = 3$, $d = 1$, and $p = 1$.

16c) The graph has asymptotes $x = -4$ and $y = x^2 - x + 5$, and y-intercept $\frac{21}{4}$. Thus $y = x^2 - x + 5 + \dfrac{1}{x+4}$, so $y = \dfrac{x^3 + 3x^2 + x + 21}{x+4}$. Hence $a = 1$, $b = 3$, $c = 1$, $d = 21$, and $p = 4$.

16d) The graph has asymptotes $x = 2$ and $y = -x^2 + x$, and y-intercept -1. Thus $y = -x^2 + x + \dfrac{2}{x-2}$, so $y = \dfrac{-x^3 + 3x^2 - 2x + 2}{x-2}$. Hence $a = -1$, $b = 3$, $c = -2$, $d = 2$, and $p = -2$.

17a)

Polynomial long division:

$$\begin{aligned} ax^3 - \left(ax^2\right)(x+1) &= -ax^2, \\ -ax^2 - (-ax)(x+1) &= ax, \\ ax - (a)(x+1) &= -a. \end{aligned}$$

$\dfrac{ax^3}{x+1} = ax^2 - ax + a - \dfrac{a}{x+1}$; asymptotes $x = -1$, $y = ax^2 - ax + a$.

17b)

Polynomial long division:

$$\begin{aligned} ax^3 - \left(ax^2\right)(x-2) &= 2ax^2, \\ 2ax^2 - (2ax)(x-2) &= 4ax, \\ 4ax - (4a)(x-2) &= 8a. \end{aligned}$$

$\dfrac{ax^3}{x-2} = ax^2 + 2ax + 4a + \dfrac{8a}{x-2}$; asymptotes $x = 2$, $y = ax^2 + 2ax + 4a$.

17c)

Polynomial long division:

$$\begin{aligned} ax^3 + bx^2 - \left(ax^2\right)(x-1) &= ax^2 + bx^2, \\ ax^2 + bx^2 - (ax + bx)(x-1) &= ax + bx, \\ ax + bx - (a+b)(x-1) &= a + b. \end{aligned}$$

The printed solution uses denominator $x - 1$: $\dfrac{ax^3 + bx^2}{x-1} = ax^2 + (a+b)x + (a+b) + \dfrac{a+b}{x-1}$. The asymptotes are $x = 1$ and $y = ax^2 + (a+b)x + (a+b)$. The question scan appears to print denominator $x + 1$.

17d)

Polynomial long division:

$$\begin{aligned} ax^3 + bx - \left(ax^2\right)(a+x) &= -a^2x^2 + bx, \\ -a^2x^2 + bx - \left(-a^2x\right)(a+x) &= a^3x + bx, \\ a^3x + bx - \left(a^3 + b\right)(a+x) &= -a^4 - ab. \end{aligned}$$

The printed solution uses denominator $x+a$: $\dfrac{ax^3+bx}{x+a} = ax^2 - a^2x + (a^3+b) - \dfrac{a^4+ab}{x+a}$. The asymptotes are $x=-a$ and $y = ax^2 - a^2x + a^3 + b$. The question scan appears to print denominator x^2+a.

17e)

Polynomial long division:

$$\begin{aligned} ax^3 - \left(\frac{ax^2}{b}\right)(bx-c) &= \frac{acx^2}{b}, \\ \frac{acx^2}{b} - \left(\frac{acx}{b^2}\right)(bx-c) &= \frac{ac^2x}{b^2}, \\ \frac{ac^2x}{b^2} - \left(\frac{ac^2}{b^3}\right)(bx-c) &= \frac{ac^3}{b^3}. \end{aligned}$$

$\dfrac{ax^3}{bx-c} = \dfrac{a}{b}x^2 + \dfrac{ac}{b^2}x + \dfrac{ac^2}{b^3} + \dfrac{ac^3}{b^3(bx-c)}$; asymptotes $x = \dfrac{c}{b}$ and $y = \dfrac{a}{b}x^2 + \dfrac{ac}{b^2}x + \dfrac{ac^2}{b^3}$.

18a)

Polynomial long division:

$$\begin{aligned} kx^2 + x^3 - 2 - \left(x^2\right)(x-1) &= kx^2 + x^2 - 2, \\ kx^2 + x^2 - 2 - (kx+x)(x-1) &= kx + x - 2, \\ kx + x - 2 - (k+1)(x-1) &= k - 1. \end{aligned}$$

Division gives $y = x^2 + (k+1)x + (k+1) + \dfrac{k-1}{x-1}$. The other asymptote is $x=1$.

18b) Comparing $x^2 + (k+1)x + (k+1)$ with $x^2 + 5x + 5$ gives $k = 4$.

19a)

Polynomial long division:

$$\begin{aligned} -kx^2 + 3kx - 9k + x^3 - 3x^2 + 9x + 20 - \left(x^2\right)(x+3) &= -kx^2 + 3kx - 9k - 6x^2 + 9x + 20, \\ -kx^2 + 3kx - 9k - 6x^2 + 9x + 20 - (-kx - 6x)(x+3) &= 6kx - 9k + 27x + 20, \\ 6kx - 9k + 27x + 20 - (6k+27)(x+3) &= -27k - 61. \end{aligned}$$

Division gives $y = x^2 - (k+3)x + (3k+9) + \dfrac{20-9k}{x+3}$. The other asymptote is $x=-3$.

19b) Comparing with $y = x^2 - 5x + 15$ gives $k = 2$.